

Seoyul Oh

Ph.D. Candidate at the University of Illinois Urbana-Champaign (UIUC)

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Interests

My research focuses on AI for networked systems, including reinforcement learning for system optimization and AI agents for reliable infrastructure management. I also work on decentralized satellite networks.

Education

University of Illinois Urbana-Champaign, IL, USA

Sep 2023 – Present

Doctor of Philosophy, Computer Science

Advisors: [Prof. Francis Y. Yan](#) & [Prof. Deepak Vasisht](#)

Korea University, Seoul, Korea

Feb 2023

Master of Science, Electrical and Computer Engineering

Advisor: [Prof. Sangheon Pack](#)

Korea University, Seoul, Korea

Feb 2021

Bachelor of Science, Electrical Engineering

Employment

Apple | Connectivity, Cellular & Satellite Team [Cupertino, CA, USA](#) 🌐

Software Engineering Intern

May 2026 – Aug 2026

Developed network optimizations for reliable real-time video streaming over bandwidth-constrained and highly variable

Direct-to-Cell LEO satellite links

Publications

- [1] **Rediscovering Heuristics: A Litmus Test for RL in Networked Systems** 📄
Seoyul Oh, Yikai Zheng, Eric Xiang Gao, Deepak Vasisht, Francis Y. Yan
ACM Workshop on Hot Topics in Networks HotNets 2026
- [2] **Exploiting Answer-Invariant Redundancies in Satellite Imagery for Efficient VLM Inference on Edge** 📄
Ishani Janveja, Davis Zhang, **Seoyul Oh**, Deepak Vasisht
NeurIPS Workshop on Machine Learning for Systems ML for Systems @ NeurIPS 2026
- [3] **EcoCell: Energy-aware Traffic Shaping for Cellular Radio Access Networks** 📄
Zikun Liu, **Seoyul Oh**, Bill Tao, Anuj Kalia, Yaxiong Xie, Deepak Vasisht
New Ideas in Networked Systems NINeS 2026
- [4] **A Call for Decentralized Satellite Networks** 📄
Seoyul Oh, Deepak Vasisht
ACM Workshop on Hot Topics in Networks HotNets 2024
- [5] **OrbitShare: A Multi-Party Architecture for Decentralized LEO Satellite Networks** 📄
Seoyul Oh, Bill Tao, Shreya Kannan, Deepak Vasisht
Under Review
- [6] **Towards Autonomous Incident Mitigation with Graph-Grounded Agents** 📄
Seoyul Oh, Nikhil Chandrasekar, Yiqi Mei, Deepak Vasisht, Francis Y. Yan
In Preparation
- [7] **An Efficient Winner and Payment Determination Algorithm in Reverse Auction for Edge Federation** 📄
Taeyun Kim, Joonwoo Kim, Hochan Lee, **Seoyul Oh**, Sangheon Pack
IEEE Transactions on Network and Service Management IEEE TNSM
- [8] **BACKWARD: A Victim-Centric DDoS Detection and Mitigation Scheme in Programmable Data Plane** 📄
Seoyul Oh, Sol Han, Hochan Lee, Sangheon Pack
IEEE Consumer Communications & Networking Conference (Poster) CCNC 2023

Research Experience

- Illinois Networked Systems and AI (INSAI) Lab, University of Illinois Urbana-Champaign** *IL, USA* Jan 2025 – Present
Research Assistant
Reinforcement Learning for Networked Systems
• Developed a framework for understanding why RL training succeeds or fails by testing whether simple policies can rediscover known heuristics and visualizing their learning behavior [1]
Agentic Infrastructure Management
• Developed graph-grounded approaches for reliable infrastructure diagnosis and incident mitigation [6]
- Wonderful Wireless Lab, University of Illinois Urbana-Champaign** *IL, USA* Sep 2023 – Present
Research Assistant
Federated Satellite Networks
• Designed decentralized architectures that enable multiple LEO satellite operators to share spare capacity for robust coverage without independently deploying large constellations [4] [5]
Energy-Efficient Edge Systems
• Developed energy-efficient systems through cellular traffic shaping [3] and redundancy-aware VLM inference for satellite imagery at the edge [2]
- Mobile System, Security and Networking Lab, Purdue University** *West Lafayette, IN, USA* Aug 2022 – Feb 2023
Visiting Graduate Researcher (Advisor: Prof. Chunyi Peng)
• Researched 5G mobility and connectivity through measurements using cellular-connected drones
- Mobile Network and Communications Lab, Korea University** *Seoul, South Korea* Mar 2021 – Jul 2022
Research Assistant
• Studied programmable data planes, network security, and resource allocation for federated edge-cloud systems

Honors and Awards

- International Fellowship** *American Association of University Women (AAUW) (25,000 USD)* Fall 2026
Korean Honor Scholarship *Ministry of Foreign Affairs, Korea (1,500 USD)* Fall 2025
KASF-KIA Scholarship *Korean American Scholarship Foundation (2,500 USD)* Fall 2025
International R&D Program Grant *Ministry of Science and ICT, Korea (12,580 USD)* Aug 2022 – Feb 2023

Teaching Experience

- CS 538: Advanced Computer Networks, University of Illinois Urbana-Champaign** *IL, USA* Spring, Fall 2025
Teaching Assistant
• Outstanding Teaching Assistant Award (Spring 2025)

Academic Service

- Artifact Evaluation Committee** ACM SIGCOMM '25, '26
Web Chair ACM SOSP PACMI Workshop '25

References

- Prof. Francis Y. Yan Assistant Professor in Computer Science at UIUC [CV]
• Prof. Deepak Vasisht Assistant Professor in Computer Science at UIUC [CV]
• Prof. Chunyi Peng Associate Professor in Computer Science at Purdue University [CV]
• Prof. Sangheon Pack Professor in Electrical Engineering at Korea University [CV]